

Major Project – Phase I(EE356)

AI-Enhanced IoT-Based Dual-Axis Solar Tracking System and Real-Time Monitoring

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Introduction

- Solar energy is a clean and renewable source of power.
- Fixed solar panels cannot always face the sun, reducing energy generation.
- Dual-axis solar tracking helps the panel follow the sun and improve efficiency.
- MPPT is used to extract maximum power from the solar panel.
- IoT enables real-time monitoring of system parameters.
- AI and Generative AI help optimize performance, analyze data, and generate intelligent reports.
- The proposed system aims to improve solar energy utilization and monitoring efficiency.

Literature review

Title of the paper	Author & Year of publication	Outcomes	Limitations
Cloud-Driven IoT Solar Tracking System with Real-Time Optimization	Multiple authors, 2026	Real-time cloud-based tracking improves solar alignment accuracy. Enhances energy efficiency using remote monitoring and adaptive control	Depends on internet reliability. Latency affects real-time response

Literature review

Dual-Axis IoT Solar Tracker with Remote Monitoring	Yonis M. Yonis Buswig et al., 2026	~21.97% efficiency improvement over single-axis systems. IoT-based alerts & remote control via cloud	Higher mechanical complexity. Maintenance cost increases
IoT-Based Dual Axis Solar Tracking System	Vinodhkumar et al., 2020 (widely cited IEEE model)	Up to 81% higher efficiency than fixed panel. Accurate tracking using azimuth & elevation angles	Requires precise calibration. Weather conditions reduce accuracy

Literature review

Comparative Study of MPPT Algorithms for PV Systems	A. Hudec et al., 2026	AI-based MPPT (ANN, LSTM, ANFIS) gives highest efficiency. Reduces oscillation and power loss	High computational complexity. Requires training datasets
Intelligent AI-Based MPPT for Solar PV Systems	M. Değermenci et al., 2026	AI models outperform traditional P&O and INC methods. Better performance under partial shading	Requires high processing hardware. Complex implementation.

Literature review

Low-Cost IoT-Enabled Dual-Axis Solar Tracking System	Springer Research Team, 2026	Achieves 40–45% higher energy output than fixed panels. Efficient dual- axis tracking using LDR sensors	LDR sensors affected by shading & noise. Low-cost controllers limit scalability
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Existing system

In most existing systems, solar panels are improved using dual-axis tracking and simple MPPT methods to get more power from sunlight. These setups often include IoT features, so users can monitor performance remotely through mobile apps or cloud platforms. While they do increase energy output compared to fixed panels, they still depend on basic sensors and conventional control techniques, which makes them less effective when weather conditions change or when there is partial shading.

Drawbacks:

- Strong dependence on internet connectivity and basic sensors, which can affect reliability.
- Limited efficiency in real-world conditions due to the use of simple MPPT methods instead of algorithms.

Problem statement & Objectives

Problem statement:

Current solar systems cannot utilize full sunlight due to fixed panels, basic tracking, and simple MPPT methods. They also struggle under changing weather and partial shading, and often depend too much on internet connectivity. This leads to reduced efficiency and unreliable performance.

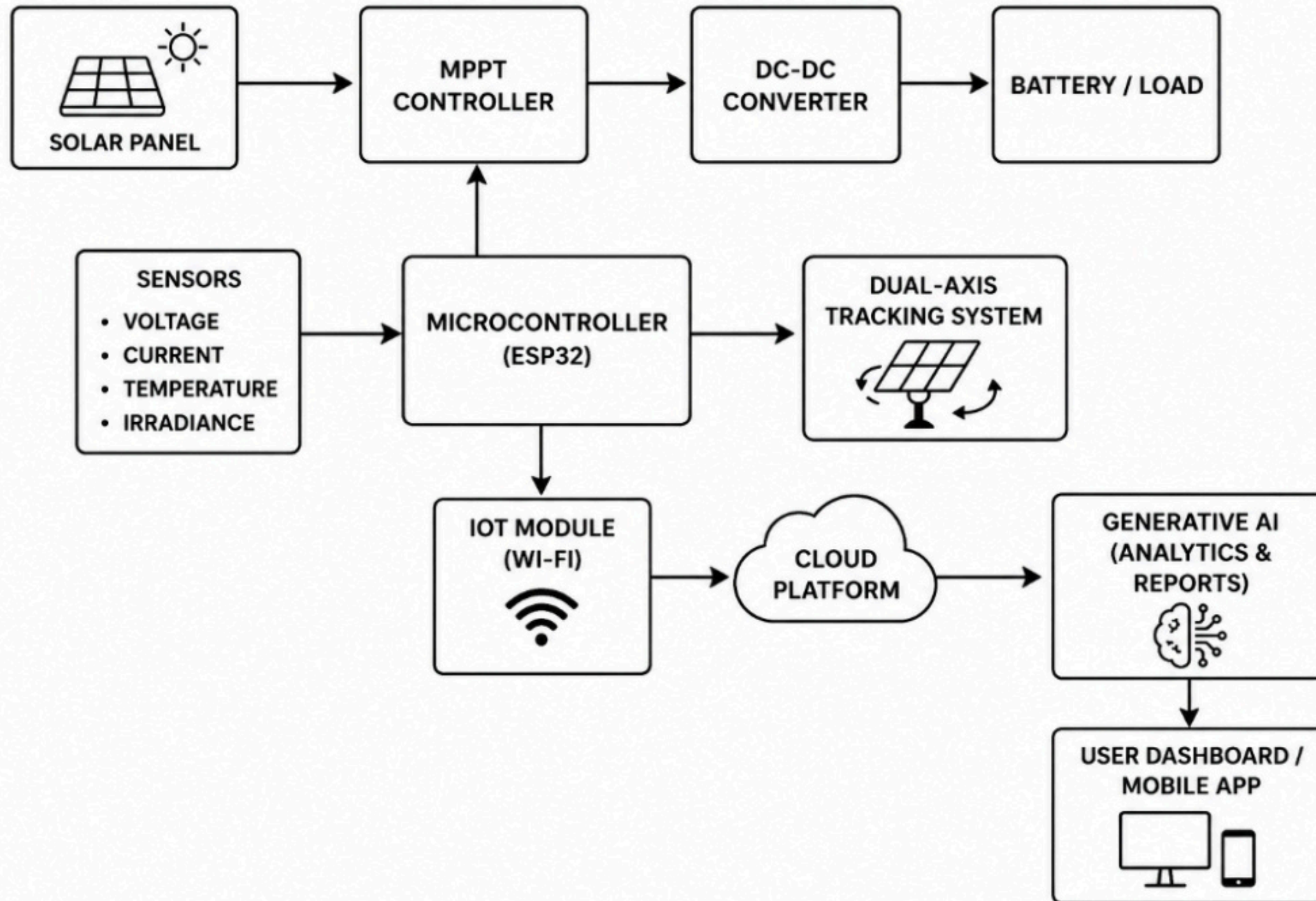
Objectives:

- To develop a dual-axis tracking system for maximum sunlight capture.
- To implement MPPT for better power extraction.
- To enable IoT-based real-time monitoring.
- To add generative AI for smart optimization.
- To make the system efficient, reliable, and less dependent on internet.

Proposed system

- Uses a dual-axis solar tracking mechanism to continuously align panels with the sun for maximum energy capture.
- Applies MPPT with AI support to extract maximum power even during partial shading and changing weather.
- Integrates IoT-based real-time monitoring for remote access and performance tracking. Supports a offline–cloud system, reducing dependency on continuous internet. Designed to be efficient, reliable, and scalable for real-world applications.

Block Diagram



Hardware and Software specification

Hardware requirements:

- Solar Panel
- MPPT Controller
- Battery
- Microcontroller (Arduino / ESP32)
- LDR Sensors
- Voltage Sensor (INA219)
- Current Sensor (ACS712)
- Motor Driver (L298N)
- DC / Servo Motors
- WiFi Module (ESP8266 / ESP32)
- Mechanical Structure
- Connecting Wires
- Power Supply

Software requirements:

- Arduino IDE
- Embedded C / C++
- ThingSpeak
- Blynk
- MATLAB
- Python
- Proteus
- Tinkercad
- Mobile App / Web Dashboard

Advantages and Applications

Advantages:

- Higher efficiency
- Maximum power output (MPPT)
- Better sunlight tracking (dual-axis)
- Real-time monitoring (IoT)
- Reduced energy loss

Applications:

- Solar power plants
- Smart homes
- EV charging
- Agriculture systems
- Street lighting
- Rural electrification

Result

The proposed system shows a clear improvement in solar energy generation compared to conventional methods. By combining dual-axis tracking with MPPT and IoT monitoring, the system is able to capture more sunlight and extract maximum power throughout the day. It performs better even under changing weather conditions, providing stable and efficient output, while also allowing users to monitor performance easily in real time.

Conclusion and future Scope

The project successfully demonstrates an efficient and intelligent solar energy system by combining dual-axis tracking, MPPT, and IoT monitoring, resulting in improved power output and better reliability compared to conventional systems. It shows that smart control and real-time monitoring can significantly enhance solar performance. In the future, the system can be further improved by integrating more advanced AI algorithms, battery management systems, and smart grid connectivity, making it more scalable, cost-effective, and suitable for large-scale real-world applications.

References

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- [6] M. Değermenci et al., “AI-Based MPPT Technique for Solar PV Systems under Partial Shading,” 2026.
- [7] A. Rihani et al., “MPPT-Based Solar PV Efficiency Enhancement Study,” 2022.
- [8] “Cloud-Driven IoT Solar Tracking System with Real-Time Optimization,” 2026.

Progress work

- Formation of the team.
- Selection of the topic.
- Discussion among the group members.
- Reporting to the guide.
- Research about the topic in online and from books.
- Preparation of ppt.
- First review

THANK YOU